PANJAB UNIVERSITY CHANDIGARH- 160014 (INDIA)

(Estted. under the Panjab University Act VII of 1947-enacted by the Govt. of India)



FACULTY OF SCIENCE

SYLLABI
FOR

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System (SEMESTER SYSTEM)

EXAMINATIONS 2024- 2025

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M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System

COURSE STRUCTURE

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System

The M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System is a two-year course, divided into four-semesters with a total of 80 credits. A student is required to complete 80 credits for the completion of the course and the award of degree.

In general, 4 hours theory lectures is equivalent to 4 Credits and 4 hours practical is equivalent to 2 credits.

Subjects offered in the M.Sc. Course are divided into four categories as follows:

(i) **Core Course (CC)** means a course that is Compulsory for a particular programme and offered by the Department, where the student is admitted.

(ii) **Discipline Specific Elective (DSE)** Course means an optional course to be selected by a student out of such courses offered by the Department, where the student is admitted.

(iii)**Ability Enhancement Course (AEC)** means an optional course to be selected by a student out of such courses offered by the Department where the student is admitted.

(iv)**Generic Elective (GE**) means an elective course which is taken by the students in the department other than where the student is admitted.

M.Sc. Scheme of examination explicitly stating **Core Courses (CC)**, **Discipline Specific Elective (DSE), Ability Enhancement Course (AEC) and Generic Elective (GE)** courses is attached along-with, followed by detailed course syllabi.

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System

PANJAB UNIVERSITY, CHANDIGARH

Outlines of Tests, Syllabi, Credits and Courses of Reading for M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System (Two Year Degree Programme) for Session 2024 – 2025.

Paper Code	Paper Name	Course Category *	Theory/ Practical Lectures	Univ. Exam Marks	Int. Ass. Marks.	Total Credits
	FIRST YEAR					
	FIRST SEMESTER					
MDS-2401	Principles of Data Science	CC	4	80	20	4
MDS-2402	Programming in Python	GE	4	80	20	4
MDS-2403	Advance Database Systems	DSE	4	80	20	4
MDS-2404	Operating System with Linux	AEC	4	80	20	4
MDS-2405	Minor Project based on MDS-2402	GE	4	40	10	2
MDS-2406	Minor Project based on MDS-2403	DSE	4	40	10	2
MDS-2407	Minor Project based on MDS-2404	AEC	4	40	10	2
					Total	**22**
	SECOND SEMESTER					
MDS-2408	Analysis and Design of Algorithms	CC	4	80	20	4
MDS-2409	Statistical Methods for Data Science	GE	4	80	20	4
MDS-2410	Data Mining and Artificial Intelligence	AEC	4	80	20	4
MDS-2411	Big Data Analytics	DSE	4	80	20	4
MDS-2412	Minor Project based on MDS-2409	GE	4	40	10	2
MDS-2413	Minor Project based on MDS-2410	AEC	4	40	10	2
MDS-2414	Minor Project based on MDS-2411	DSE	4	40	10	2
					Total	**22**

***Core Courses (CC), Discipline Specific Elective (DSE), Ability Enhancement Course (AEC) and Generic Elective (GE).**

****From SWAYAM or among the courses being offered by other departments of the University. Students can complete any MOOC course of minimum 12 weeks from SWAYAM.**

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GUIDELINES FOR SUBMISSION OF PROJECT REPORT (MDS - 2421)

The report should consist of the following:

- Cover page including Project title, Name of the student, Name of the Department and Names of the Project Guides (both External and Internal).
- Acknowledgements.
- Certificates from company and department duly signed by external guide, Principal and internal guide.
- Contents with page numbers.
- Introduction (includes background and application or importance of the project)
- Objectives
- System Analysis

System Feasibility study

- Software requirement specifications
- Design with system flowcharts and input/output design.
- Implementation and Testing

- Hardware and software used
- Listing of well commented programs with result/output or detailed algorithms with input and output.

Further scope of the project

- Bibliography
- Appendices (any other information related to project)

Each student should observe the following norms while submitting the synopsis/thesis for the Project:

(a) Use both sides of the paper instead of only single side.
(b) Use one and half interline spacing in the text (instead of double space)
(c) Stop using a blank sheet before the page, carrying figure or table.
(d) Try to insert figure/table in the text page itself (instead of using a fresh page for it, each time.)

Students must consult/inform the internal guides regarding the progress of their work at least once in 20 days. It is the duty of the student to be in touch with his/her internal guide. The student must prepare 3 copies of the report including one copy for self. The remaining two copies are to be submitted before 31st May every year as per the following:

1. Department Library
2. Internal Guide

One softcopy of the work is to be submitted to the concerned head of the department/institution along with the report. The student must present his/ her work in 10 - 15 minutes mainly focusing on his/her contribution with the help of slides followed by demonstration of the practical work done. The project Viva will be completed before 15th June every year exact dates will be informed before 31st May every year.

An external examiner, internal examiner and the internal guide will conduct project viva.

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SYLLABUS AND COURSE OF READINGS
FIRST SEMESTER

Paper Title: Principles of Data Science

Paper Code: MDS - 2401 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives: The main objectives of this course are to:

- provide a strong foundation for data science and its application area.
- understand the underlying core concepts and emerging technologies in data science.
- develop applied experience with data science software, programming, applications and processes.
- develop practical skills needed in modern analytics.
- give a hands-on experience with real-world data analysis.

Note:

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes: After successful completion of the course, student will be able to:

- Understand the impact of digital data in businesses its role in making crucial decisions.
- Have preliminary knowledge on data science process, tools, and job trends.
- Have proficiency in utilizing diverse feature selection, anomaly detection methods and time series forecasting techniques.
- Have knowledge about available data science tools.

UNIT I

Data Science: History, Principles and Impact; Data: Types, Modelling, Exploring, Collecting, Visualizing, Predicting, and Deploying the Model; Data Science Process, Lifecycle; Components of Data Science: Data Analysis process and types.

UNIT II

Data Science Integrating Statistics, Computer Science, and Domain Expertise: Statistics and Mathematical Functions; Data Science tools: Database, Data Analytics, Web Scraping, Machine Learning, Reporting. Data Science as a profession: Skill set and Job Trends.

UNIT III

Feature selection and Forecasting: Introduction, Feature Selection: Classifying feature selection methods; Anomaly Detection: Introduction, Distance and Density based outlier detection, Local Outlier Factor, Time series Forecasting, Decomposition, Smoothing based methods, Regression based methods, Machine Learning methods.

UNIT IV

Data science tools and applications: Introduction to Data Science Tools: SAS, APACHE FLINK, BigML, Excel, Tableau, Matplotlib, TensorFlow, Weka. Applications: Solving Data Problems: Collecting and Analyzing Social media Data.

Suggested Readings:

1. Joel Grus: Data Science from Scratch – First Principles with Python, O'Reilly Media, SPD), 2017.
2. Foster Provost and Tom Fawcett: Data Science for Business – What you need to know about Data Mining and Data-Analytic Thinking, O'Reilly Media, SPD, 2019.
3. Peter Flach: Machine Learning - The Art and Science of Algorithms that Make Sense of Data, Cambridge University Press, 2015.
4. Davie Cielen, Arno D.B. Meysman, and Mohamed Ali: Introducing Data Science – Big Data, Machine Learning and More, Using Python Tools, Dreamtech Press, 2016.
5. Sanjeev J. Wagh, Manisha S. Bhende, Anuradha D. Thakare, Fundamentals of Data Science, 1st Edition, 2022.
6. Daimi, Kevin, Ed. Hamid R. Arabnia, Principles of Data Science, Springer, 2020.
7. Vijay Kotu, Bala Deshpande, Data Science: Concepts and Practices, Morgan Kaufmann Publishers, Second edition, 2019.
8. D J Patil, Hilary Mason, Mike Loukides, Ethics and Data Science, O' Reilly, 1st edition, 2018.
9. Sinan Ozdemir, Principles of Data Science, Packt Publishing, 2016.

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Paper Title: Programming in Python

Paper Code: MDS - 2402 **Max. Marks: 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives: The main objectives of this course are to:

- To provide an overview of Python Programming
- To identify and explain basic Python syntax, including variables, data types, and operators.
- To teach the fundamental techniques and principles in decision making and iteration using Python.
- To develop algorithmic thinking skills to solve problems using Python.
- Apply Python programming concepts to devise solutions for real-world problems.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii)The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes: After successful completion of the course, student will be able to:

- Demonstrate Python Programming syntax, including variables, data types, and operators, enabling them to write clear and concise Python code.
- Develop strong problem-solving skills by applying control flow structures and functions to design solutions for a variety of computational problems.
- Use fundamental data structures in Python, such as lists, tuples, dictionaries, and sets, and will be able to manipulate and analyze data effectively.
- Acquire a solid foundation in object-oriented programming (OOP), understanding how to create and use classes, objects, inheritance, and polymorphism in Python for building modular and scalable programs.

UNIT I

Introduction to Object Oriented Programming (OOPs), OOPs Concepts-Class, Object, Encapsulation, Abstraction, Polymorphism, Message Passing, Static and Dynamic Binding.

Introduction to Python, History of Python, Python Interpreter, Installation of Python in Windows and Linux, Python IDE, Introduction to Anaconda, Basic Program Execution in Python, Python Shell.

UNIT II

Python Programming Basics- Variable Declaration, Keywords, Indents and basic Input-output operations in Python. Operators in Python : Arithmetic Operators, Assignment Operators, Comparison

Operators, Logical Operators, Bitwise Operators, Identity Operators, Membership Operators, Ternary Operator, Operator precedence.

Data Types in Python : Built-in, String, List, Tuple, Set, Dictionary, Basic list operators, list slicing, replacing, inserting, removing an element, Searching, Sorting; Dictionary literals: adding & removing keys, accessing & replacing values, traversing dictionaries; String Manipulations: Subscript operator, indexing, slicing a string, other functions on strings, string module. Strings and number system: Format functions, converting strings to numbers and vice versa.

UNIT III

Conditional Statements and Loop Conditional Statements in Python: If, If-else, If-elif-else, Nested-if and for, while, Nested loops, Break, Continue, Pass statements.

Introduction to functions in Python, Function definition and calling, Function parameters, Default argument function, Variable argument function, built-in functions in Python, Scope of variable. Recursive functions, Global statements, Importing modules, Math modules and Random modules.

UNIT IV

File Handling in Python: Introduction to Data, Records, Fields, Files, Attributes of a File, File opening modes, closing of a file, Reading from a file, Writing onto a file, some important File handling functions open(), close(), read(), readline().

Arrays in Python, Concept of modularization, Importance of modules in Python, Importing modules, using built-in modules like Numpy.

Suggested ReadingS :

1. Wesley J. Chun : Core Python Applications Programming, 3rd Edition, Pearson Education, 2016.
2. Charles Dierbach : Introduction to Computer Science using Python, Wiley, 2015.
3. Downey, A. : How to think like a Computer Scientist: Learning with Python, John Wiley, 2015.
4. Mark Lutz : Learning Python, 5th edition, Oreilly Publication, 2013.
5. John Zelle : Python Programming An Introduction to Computer Science, Second edition, Course Technology Cengage Learning Publications, 2013.
6. Michel Dawson: Python Programming for Absolute Beginners, Third Edition, Course Technology Cengage Learning Publications, 2013.
7. David Beazley, Brian Jones : Python Cookbook, Third Edition, O'Reilly Publication, 2013.

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Paper Title: Advance Database Systems

Paper Code: MDS - 2403 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives:

- Develop a deep understanding of advanced database systems, covering topics such as distributed databases, data warehousing, and No SQL databases.
- Acquire hands-on experience in designing and implementing complex database systems, including optimization techniques and performance tuning.
- Stay abreast of the latest trends in database technology, exploring topics like blockchain databases, graph databases, and in-memory databases.
- Gain expertise in advanced database security measures and privacy considerations, ensuring the ability to design and implement secure and compliant data systems.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes: After completing this course the students will be able to:

- Demonstrate proficiency in advanced database design, optimization, and implementation, showcasing their ability to architect robust and scalable data systems.
- Apply their knowledge to implement and manage databases using emerging technologies, such as NoSQL, distributed databases, and in-memory databases.
- Exhibit leadership in data security and compliance, implementing advanced measures to safeguard sensitive information in accordance with industry standards and regulations.
- Possess strategic data management skills, ensuring their ability to handle large-scale datasets efficiently.

UNIT I

Database Systems: DBMS: Components, Views of data-schemas and instances, Data independence, 3-tier Architecture, Data models classification: File based System, Traditional data models - Hierarchical, Network, Relational Models. Entity- relationship model, Representation of relationship set, Generalization, Aggregation, Normalization.

UNIT II

Relational Database Systems: Relational Database: Attributes, Domains, Tuples, Relations and their schemes. Relational Algebra: Operations- union, intersection, difference, Cartesian product, projection,

selection, division and relational algebra queries; Relational Calculus: Tuple oriented and domain oriented relational calculus and its operations. Introduction to PL/SQL: block structure, Architecture, SQL within PL/SQL, Writing PL/SQL code.

UNIT III

Transaction Management and Concurrency control: Concept of Transaction, ACID properties, Serializibility, States of transaction, Concurrency control: Locking techniques, Time stamp based protocols, Granularity of data items, Deadlock.
Query Optimization: Query execution plans, Cost-based optimization, Techniques for improving query performance.

UNIT IV

Temporal and Spatial Databases: Introduction to Temporal Database: Time ontology, structure, and granularity, Temporal data models. Introduction to Spatial Database: Definition, Types of spatial data, Geographical Information Systems (GIS).
NoSQL Databases: Overview of NoSQL databases (document-oriented, key-value, column-family, graph), Use cases and suitability of NoSQL databases, Implementation and comparison of popular NoSQL databases (MongoDB, Cassandra, Redis)

Suggested Readings:

1. C.J.Date : Introduction to Database Systems.
2. Elmasri Navathe : Fundamentals of Database Systems.
3. Ivan Bayross : PL/SQL The Programming Language of ORACLE, BPB Publication.
4. Vijay Mukhi: Mastering Oracle 6.0, BPB Publications.
5. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan: Database System Concepts.
6. Christian S. Jensen and Richard T. Snodgrass: Temporal Database Management: A Survey.
7. Shashi Shekhar and Sanjay Chawla : Spatial Databases: A Tour.

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Paper Title: Operating System with Linux

Paper Code: MDS - 2404 **Max. Marks : 80 Time : 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives: To understand the various concepts of Operating System like process management, synchronization, deadlocks, storage and memory management.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes: After successful completion of course the students will be able to:

- Describe the important computer system resources and the role of operating system in their management policies and algorithms.
- Understand the process management policies and scheduling of processes by CPU
- Evaluate the requirement for process synchronization and coordination, memory management handled by operating system also cater-policies with respect to different storage.
- Learn to use features of Linux for programming and administration

UNIT I

Introduction to Operating System: Introduction to Operating System, Its need and services; Different types of operating systems: batch, multi-programmed, time sharing, real time, distributed, parallel.

Process Management and Synchronization: Process: Process state, Process control block, Inter process communication: Shared memory systems, Message passing systems; CPU Scheduling-scheduling criteria, Preemptive & non-pre-emptive scheduling, Scheduling Algorithms: FCFS, SJF, RR and Priority.

Synchronization: Critical section problem, Peterson's solution, Synchronization hardware, Semaphores: Mutual exclusion, Producer-consumers, Reader-writers problem; Dining philosophers Problem;

UNIT II

Deadlocks: Introduction to deadlocks, Conditions for deadlock, Resource allocation graphs, Deadlock Detection and Recovery, Deadlock Avoidance, Deadlock Prevention

Memory Management and Disk Scheduling: Logical vs. Physical address space, Memory management without swapping; Swapping; Contiguous memory allocation, Paging, Segmentation, Segmentation with paging; Virtual Memory, Demand paging, Page replacement algorithms (FIFO, Optimal, LRU); Thrashing.

Disk Scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, and LOOK), Disk Management.

UNIT III

Features of LINUX Operating System: Linux File System - ext2 and ext3 file system architecture, blocks and fragments, I-node directory structure; File and Directory management features; Process management features, Text manipulation features; Regular expressions, Features of Bourne Again Shell (BASH).

Linux Shell Scripting: Programming in the Bourne Again Shell (BASH); Wild card characters, Simple shell scripts; Shell variables; shell programming constructs; interactive shell scripts; Advanced features.

UNIT IV

Linux System Administration: Definition, Role and functions of a system administrator, Booting the system, Run Levels in Linux, Maintaining user accounts, File System and Special Files, Backups and restoration.

Linux System Calls: C as System Programming Language; I/O system calls – umask(); create(); open(); read(); write(); lseek(); dup(); link(); access(); chmod(); chown(); Process management system calls – fork(); getpid(); getppid(); exit(); wait(); sleep() ; Signal system calls – kill(); signal(); Exec System Calls – execl();execlp();execv();execvp().

Suggested Readings:

1. Peterson, James, L. and Silberschatz, A. : Operating System Concepts, Wiley Publ. Comp, 1985.
2. Dhamdhere,D M : Operating Systems-A concept based approach,Mc Graw Hill.
3. Deitel, H.M.: An Introduction to Operating System, Addison-Wesley Publ.Comp, 1984.
4. Milenkovic, M.: Operating System – Concepts and Design, McGraw Hill, International Editions, 1987.
5. Richie: Operating System, BPB.
6. Hansen PerBrineh : Operating System Principles, Prentice Hall India, 1978.
7. Madnick and Donovan: Operating System, McGraw Hill Book Co.
8. Joshi, R.C.: Operating Systems, Wiley India Pvt. Ltd.
9. Parker, Tim: Linux Unleashed, Latest Edition, Techmedia.
10. Stones, Richard and Mathew Neil : Beginning Linux Programming, 3rd Edition, Wrox.

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PRACTICAL EXAMINATIONS (SESSION 2024 - 2025)
The Practical examination will be conducted for each of the following:

Paper Code: MDS - 2405 (Minor Project Based on MDS - 2402)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.

Paper Code: MDS - 2406 (Minor Project Based on MDS - 2403)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.

Paper Code : MDS - 2407 (Minor Project Based on MDS - 2404)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.

SEMESTER II

Paper Title: Analysis and Design of Algorithms

Paper Code : MDS - 2408 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: The objective of the module is to create skills in students to design and analyze algorithms.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes : After successful completion of course the students will be able to:

- Analysis of algorithms including recursive algorithms, space and time complexity.
- Understand the Divide and Conquer design strategy to solve different problems like sorting, selection and matrix multiplication.
- The students will also be able to learn various Dynamic Programming design techniques.
- Backtracking and Branch n Bound design techniques and application of these techniques to solve problems. Deal with finding lower bounds on complexity for sorting and search problems using comparison trees and the lower bounds on parallel computations. NP-Hard and NP-Complete problems and the examples of such problems able be written.

UNIT I

Algorithms and Analysis : Introduction, Algorithms specification, Recursive algorithms, space and time complexity, Asymptotic Notation (O,Ω, and Θ; o,ω), Practical complexities, Best, Average and worst case performance of algorithms, Examples, Introduction to recurrence relations: Solving recurrence relations using repeated substitutions.

Divide and Conquer: General method, Binary search, Merge sort, Quick sort, Selection problem, Matrix Multiplication using Divide and Conquer and Stassen's method and analysis of these problems.

UNIT II

Greedy Method: General Method, Knapsack problem, Job Sequencing with deadlines Problem, Minimum spanning Trees, Analysis of these problems.

Dynamic Programming: General method, Optimal binary search trees, 0/1 Knapsack, Travelling salesperson problem.

UNIT III

Back Tracking : General method, 8 queen's problem, Graph coloring, Hamiltonian cycles and Analysis of these problems.

Branch-And-Bound : General Method least cost search, upper bounding, 0/1 Knapsack and Travelling Salesperson problems, Dynamic state space organization for Travelling Salesperson problem.

UNIT IV

Lower-Bound Theory: Introduction to lower bounds, Comparison Trees, Lower bounds for ordered searching and sorting, Lower bounds on Parallel Computations: Lower bound on average case sorting.

NP-hard and NP-complete problems: Basic concepts, Deterministic and Non-Deterministic algorithms, NP-Hard and NP-Complete classes, Statement of Cook's Theorem, Examples of NP-hard graph and NP-Hard scheduling problems, NP-Hard Graph problem with proof: Clique Decision Problem.

Suggested Readings:

1. Horowitz, Ellis and Sahni, Sartaj : Fundamentals of Computer Algorithms, Universities Press Publications.
2. Aho, A.V., Hopcroft, J.E., Ullman, J.D. : The Design and Analysis of Computer Algorithms, Addison-Wesley.
3. Bentley, J.L. : Writing Efficient Programs, Prentice-Hall India, Eastern Economy Edition.
4. Goodman, S.E. & Hedetniemi : Introduction to the Design and Analysis of Algorithms, McGraw- Hill Book Comp, 2004.
5. Knuth, D. E. : Fundamental of Algorithms The Art of Computer Programming, Vol.-1, Naresh Publ. House.
6. Brassad, Gilles and Bartley, Paul : Fundamentals of Algorithms, Prentice Hall of India, 1986.
7. Mark Allen Weiss : Data Structure and Algorithms Analysis in C++, Pearson Education.

Paper Title: Statistical Methods for Data Science

Paper Code : MDS - 2409 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective: To know the fundamental concepts Statistical Methods used in Data science and to explore its practical implementation in real-life applications.

Note :
(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcomes: After successful completion of course the students will be able to:

- Conceptual understanding of R language and its understanding for statistical computing and graphics. Implementation of R fundamentals and data frames.
- Knowledge of hypothesis testing and curve fitting.
- Understanding of important industrial use concepts as regression, classification, dimension reduction and clustering.
- Understanding of important statistical based machine learning concepts as support vector machines, decision trees and random forest techniques.

UNIT I

R Language: Programming with R basic data types, vector arithmetic, matrices, factor levels, ordered factors, data frame, sub setting and sorting data frames, charts and graphs in R, linear models in R as linear regression and multiple regression. Functions in R: writing a function in R, function scoping, recursion, R Package. Handling of csv files and datasets using R.

UNIT II

Data Distribution and Hypothesis Testing: Probability and Continuous distribution, Distribution of sample data, hypothesis testing, T-test, binomial distribution, chi-square distribution, F-distribution, Poisson distribution, confidence intervals, parametric intervals

Curve Fitting: Linear regression with one predictor, least square for polynomial models, local fitting, big data cloud, visualizing multivariate data

UNIT III

Regression and Classification: Multiple linear regression, polynomial regression, logistic regression, support vector machines, k-nearest neighbor, Naïve bayes theorem, decision tree and random forest

UNIT-IV

Dimension Reduction and Clustering: Introduction to unsupervised learning, Principal component analysis, linear discriminant analysis, k-means and hierarchical clustering

Suggested Readings:

1. Walter A. Rosenkrantz : Introduction to probability and statistics for scientists and engineers, McGraw-Hill.
2. Gareth James, Daniela Witten, Trevor Hastie Robert Tibshirani : An Introduction to Statistical Learning with Applications in R, Springer.
3. K.G. Srinivas, G M Siddesh : Statistical Programming in R, Oxford Publications.
4. Stanley H. Chan : Introduction to Probability for Data Science, Michigan Publishing
5. Hadley Wickham and Garrett Grolemund, R: Data Science, O-Reilly Publishers
6. Carlos Andre Reis Pinheiro and Mike Patetta : Introduction to Statistical and Machine Learning Methods for Data Science, SAS Institute Publishers.
7. Christopher M Bishop : Pattern Recognition and Machine Learning, Springer.
8. S. Sridhar and M. Vijayalakshmi : Machine Learning, Oxford University Press.

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Paper Title: Data Mining and Artificial Intelligence

Paper Code : MDS - 2410 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objectives:

- Equip students with the theoretical foundations and practical skills in data mining and artificial intelligence
- Fostering proficiency in implementing architectural algorithms
- Exploring emerging trends and promoting ethical application across diverse domains.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course outcomes: After completing the course the student will be able to:

- Have understanding of data warehousing fundamentals and data mining techniques like association, classification, and clustering.
- Have knowledge of Artificial intelligence historical context, heuristic search methods, and knowledge representation encompassing logic and semantic networks.
- Have possess knowledge of specialized AI areas, including expert systems, natural language processing, genetic algorithms, and artificial neural networks covering various learning paradigms.

UNIT I

Data Warehousing: Data Warehousing: Definition, usage and trends, Characteristics of a Data Warehouse, DBMS vs. data warehouse.
Data Mining :Steps of Data Mining Process, Data Modeling for data warehouses, OLAP and OLTP, Association rules, Classification, Clustering, Regression.
Classification and Clustering: Dependency Modeling, Link Analysis, Sequencing Analysis, Social Network Analysis.

UNIT II

Introduction to Artificial Intelligence: Turing Test, History and Development in AI, applications of AI, State Space representation, Production Systems, characteristics and problem characteristics.
Heuristic Search Techniques: Introduction to heuristic search, Hill Climbing, game playing, min-max search procedure, reducing alternatives using Alpha-Beta pruning cutoff method.
Knowledge Representation: Knowledge representation methods, Logic: Propositional logic and first order predicate logic, Semantic networks, Frames, Rules, Scripts, Handling uncertainty in knowledge.

UNIT III

Expert Systems: Expert Systems: Characteristics and Architecture.
Natural Language Processing: Grammar and Language, Parsing Techniques, Semantic Analysis and Pragmatics, Conceptual Dependency and Ontologies.

UNIT IV

Fuzzy sets: Notion of fuzziness, Membership functions, fuzzification and defuzzification, Operations on fuzzy sets, fuzzy functions and linguistic variables.
Genetic Algorithms(GA): Encoding strategies, Genetic operators, Fitness functions and GA cycle , Problem solving using GA.
Artificial Neural Networks (ANN): Supervised, unsupervised and reinforcement Learning, Single Perceptron, Multi Layer Perceptron, Self Organizing Maps, Hopfield Network.

Suggested Readings:

1. Jiawei Han, Micheline Kamber, Jian Pei, Hanghang Tong : Data Mining: Concepts and Techniques, Third Edition, Elsevier.
2. Rich Elaine and Knight Kevin Shiva Shankar B. Nair : Artificial Intelligence, Third Edition, Tata MacGraw Hill, 2017.
3. Patterson, Dan W. : Introduction to Artificial Intelligence and Expert Systems, Pearson Education, 2015.
4. S.N. Sivanandam and S.N. Deepa : Principles of Soft Computing, Second Edition, John Wiley.
5. Daniel Jurafsky and James H. Martin : Speech and Language Processing: Second Edition, Pearson.

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System

Paper Title: Big Data Analytics

Paper Code : MDS - 2411 **Max. Marks: 80 Time: 3 Hrs.**
Course Duration: 60 Lectures of one hour each.

Objective:

- To provide an overview of an exciting growing field of big data analytics.
- To introduce the tools required to manage and analyze big data like Hadoop, MapReduce.
- To teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability.
- To enable students to have skills that will help them to solve complex real-world problems in for decision support.

Note :

(i) The Question paper will consist of four units.
(ii) Examiner will set total of **nine** questions comprising two questions from each unit and one compulsory question of short answer type covering whole syllabi and having equal distribution of marks from all the units.
(iii) The students are required to attempt one question from each unit and the compulsory question.
(iv) All questions carry equal marks.

Course Outcome: After the completion of the course the students will be able to

- Understand the key issues in big data management and its associated applications in intelligent business and scientific computing.
- Acquire fundamental enabling techniques and scalable algorithms like Hadoop, and Map Reduce in big data analytics.
- Interpret business models and scientific computing paradigms, and apply software tools for big data analytics.
- Achieve adequate perspectives of big data analytics in various applications like recommender systems, social media applications etc.

UNIT I

Introduction to Big Data - Definition, need and importance, Characteristics of Big Data and Dimensions of Scalability, Foundations for Big Data Systems and Programming, Application of Big Data Analytics.

UNIT II

Hadoop: History of Hadoop, The Hadoop Distributed File System, Components of Hadoop, Analyzing the Data with Hadoop, Design of HDFS, HDFS concepts, Hadoop I/O, data integrity, compression, and serialization, Setting up Hadoop Eco System.

UNIT III

MapReduce: Introduction, Using Elastic MapReduce, Understanding MapReduce key-value pairs, The Hadoop Java API for MapReduce, Writing MapReduce programs, Hadoop specific data types, Input and output.

UNIT IV

HIVE and PIG - Introduction of Hive, Hive Features, Hive architecture, Hive Vs. RDBMS, Hive Meta store, Hive data types, Hive Tables, Table types, Creating database, DDL and DML, Built-In Functions and Operators, User defined functions.
PIG - Overview of Pig, Pig Architecture, Pig Execution modes, Pig Grunt shell and Shell - commands. Pig Execution Modes – Batch Mode, Embedded Mode, Pig Execution in Batch Mode. Embedding Pig in Python – Use cases, Map Reduce programs with Pig, Pig Vs SQL.

Suggested Readings:

1. Chris Eaton, Dirk Deroos et al. : Understanding Big data, McGraw Hill, 2012.
2. Tom White : Hadoop: The Definitive Guide, Fourth Edition, O′Reilly Publishers, 2012.
3. Donald Miner, Adam Shook : MapReduce Design Patterns, O'Reilly Media November 22, 2012.
4. Edward Capriolo, DeanWampler, Jason Rutherglen : Programming Hive, O'Reilly Media; 1 edition, October, 2012.
5. Deepak Vohra : Practical Hadoop Ecosystem - A Definitive Guide to Hadoop-Related Frameworks and Tools First Edition, Apress Publisher, ISBN: 9781484221983, 2016
6. Alan Gates : Programming Pig, O'Reilly Media; 1st Edition, October, 2011.

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of Honours School System

PRACTICAL EXAMINATIONS (SESSION 2024 - 2025)
The Practical examination will be conducted for each of the following:

Paper Code : MDS - 2412 (Minor Project Based on MDS - 2409)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.
iii) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.

Paper Code : MDS - 2413 (Minor Project Based on MDS - 2410)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.
iii) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.

Paper Code : MDS - 2414 (Minor Project Based on MDS - 2411)

Time : 3 Hrs.
Max. Marks : 50
External : 40
Internal : 10

Note : i) The Internal Assessment Marks for each practical will be based on Minor Project.
ii) Paper will be set at the time of examination. Due weight-age may be given to the practical notebook and assignments for evaluation.
iii) Each student is required to undergo 04 to 06 weeks Summer Training at the end of Second Semester.